EWB AWS VPC PEERING PROJECT

Private Connectivity Between Two Amazon VPCs

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Batch	Batch 1
AWS Region	Asia Pacific (Mumbai) — ap-south-1
Completion date	26 October 2026

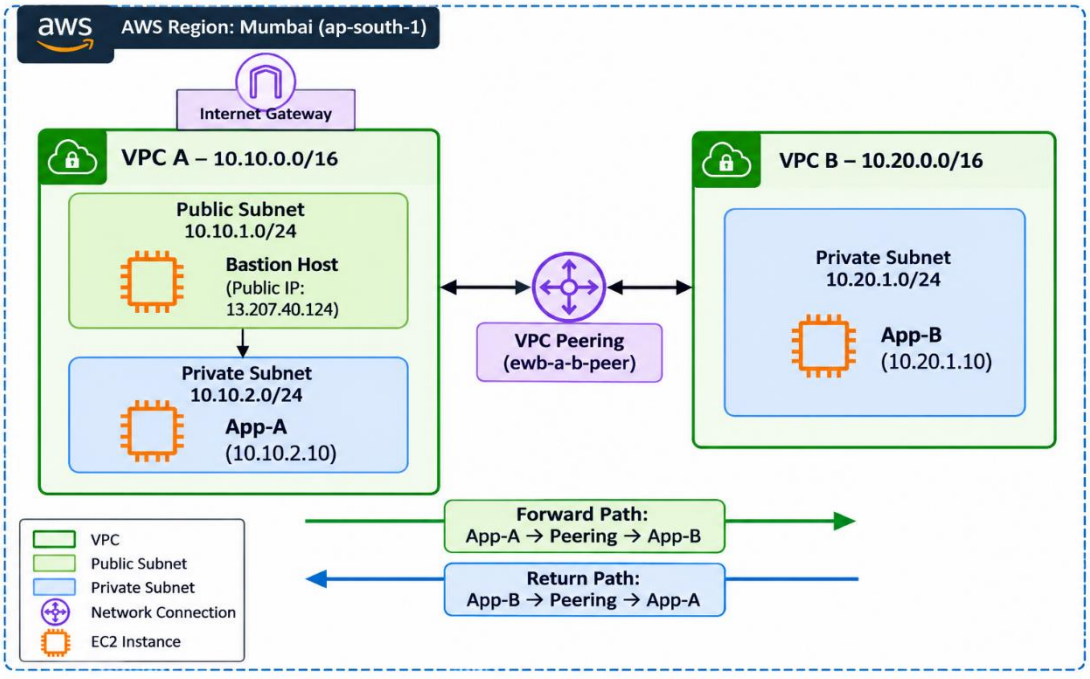
1. Project Overview

The objective of this project was to configure private network connectivity between two Amazon Virtual Private Clouds (VPCs) using VPC Peering. The lab used a bastion host in the public subnet of VPC A and private application instances in both VPCs. Connectivity was validated with ICMP ping tests in both directions and SSH access to the private App-B instance through the bastion host.

Region: Asia Pacific (Mumbai), ap-south-1
VPC A: ewb-vpc-a — 10.10.0.0/16
VPC B: ewb-vpc-b — 10.20.0.0/16
Peering connection: ewb-a-b-peer — pcx-029d206e3eaab2d89 (active before cleanup)

2. Architecture

Insert the saved architecture diagram in this section. The architecture should show VPC A (10.10.0.0/16), VPC B (10.20.0.0/16), the three subnets, the bastion host, both private application instances, the internet gateway, and the VPC peering connection.



Architecture diagram

3. Resource Inventory

Resource	Name / ID	Address / CIDR / Status
VPC A	ewb-vpc-a vpc-036668c878501fcab	10.10.0.0/16
VPC B	ewb-vpc-b vpc-07680626b64386088	10.20.0.0/16
Public subnet A	ewb-a-public	10.10.1.0/24
Private subnet A	ewb-a-private	10.10.2.0/24
Private subnet B	ewb-b-private	10.20.1.0/24
Bastion EC2	ewb-bastion i-0be13e89eb510e1f1	Public IP 13.207.40.124
App-A EC2	ewb-app-a i-025cd251f13bd4286	Private IP 10.10.2.10
App-B EC2	ewb-app-b i-0753b76acf936d7ac	Private IP 10.20.1.10
VPC Peering	ewb-a-b-peer pcx-029d206e3eaab2d89	Active before cleanup
Internet Gateway	ewb-igw-a igw-015c657761a276b3f	Deleted during cleanup
Public route table	ewb-rt-a-public rtb-0c78e1d7c268a9211	Associated with ewb-a-public
Private route table A	ewb-rt-a-private rtb-06a8df95bc85ac22d	Associated with ewb-a-private
Private route table B	ewb-rt-b-private rtb-0d6146a003fafcad4	Associated with ewb-b-private
Bastion SG	ewb-sg-bastion sg-07236205dc0bdb555	Deleted during cleanup
App-A SG	ewb-sg-app-a sg-09ac255ab63a13ffe	Deleted during cleanup
App-B SG	ewb-sg-app-b sg-00ac073f62c021b0a	Deleted during cleanup

4. Subnet and Route Configuration

The lab used the following subnet CIDRs and route-table associations:

Route table	Subnet association	Routes configured during the lab
ewb-rt-a-public	ewb-a-public (10.10.1.0/24)	10.10.0.0/16 — local 0.0.0.0/0 — Internet Gateway 10.20.0.0/16 — VPC peering
ewb-rt-a-private	ewb-a-private (10.10.2.0/24)	10.10.0.0/16 — local 10.20.0.0/16 — VPC peering
ewb-rt-b-private	ewb-b-private (10.20.1.0/24)	10.20.0.0/16 — local 10.10.0.0/16 — VPC peering

The public route table provided internet-bound routing through the internet gateway. The private route tables included routes to the other VPC CIDR through the peering connection. VPC peering is not a gateway to the internet; both VPCs require appropriate routes and security rules for private communication.

5. Security Group Rules

The recorded inbound rules were narrowly scoped to the lab hosts:

ewb-sg-bastion: SSH (TCP 22) from the user's public IP address /32. **ewb-sg-app-a:** SSH (TCP 22) from 10.10.1.10/32; ICMP Echo Request from 10.20.1.10/32. **ewb-sg-app-b:** SSH (TCP 22) from 10.10.1.10/32; ICMP Echo Request from 10.10.2.10/32.

Default outbound rules were retained.

Important: The recorded security-group source for SSH to the application instances is 10.10.1.10/32. The Bastion private IP was not recorded, so this report preserves the recorded rule rather than assuming the source address was the Bastion's private IP.

6. Connectivity Tests and Evidence

The following test results were recorded during the lab. Insert the corresponding saved screenshots and label each one with the source host, test, and result.

Baseline test: Ping between the private application hosts failed before peering/routing was fully configured. This established the initial lack of connectivity.

Forward test: App-A (10.10.2.10) pinged App-B (10.20.1.10). Four replies were received with 0% packet loss.

Reverse test: App-B (10.20.1.10) pinged App-A (10.10.2.10). Four replies were received with 0% packet loss.

Private SSH test: SSH access to App-B was demonstrated through the Bastion host; App-B had no public IPv4 address.

Controlled ICMP test: ICMP connectivity was deliberately blocked by changing the relevant rule, the failure was observed, and connectivity was restored after the rule was recovered.

Screenshot placement checklist

Figure 1: Baseline Connectivity Test Before VPC Peering Configuration

A ping test was performed between the private EC2 instances in VPC A and VPC B before the required peering connectivity was established. The test failed, demonstrating that private communication between the two VPCs was not yet working.

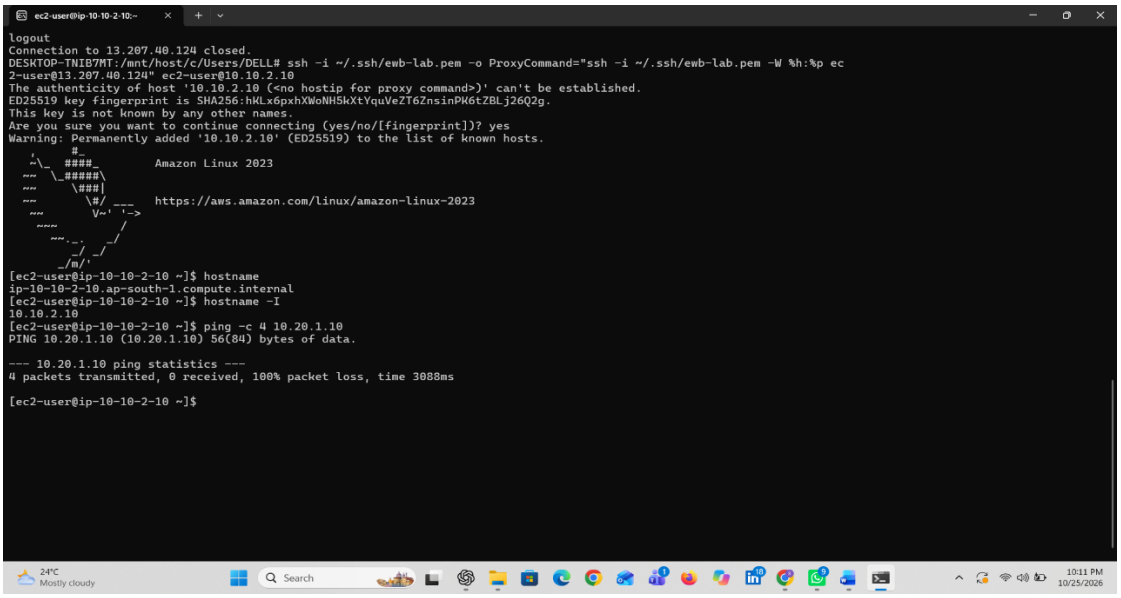


Figure 2: Active VPC Peering Connection Between VPC A and VPC B

The VPC peering connection ewb-a-b-peer (pcx-029d206e3eaab2d89) connected ewb-vpc-a (10.10.0.0/16) and ewb-vpc-b (10.20.0.0/16). The connection was active before project cleanup.

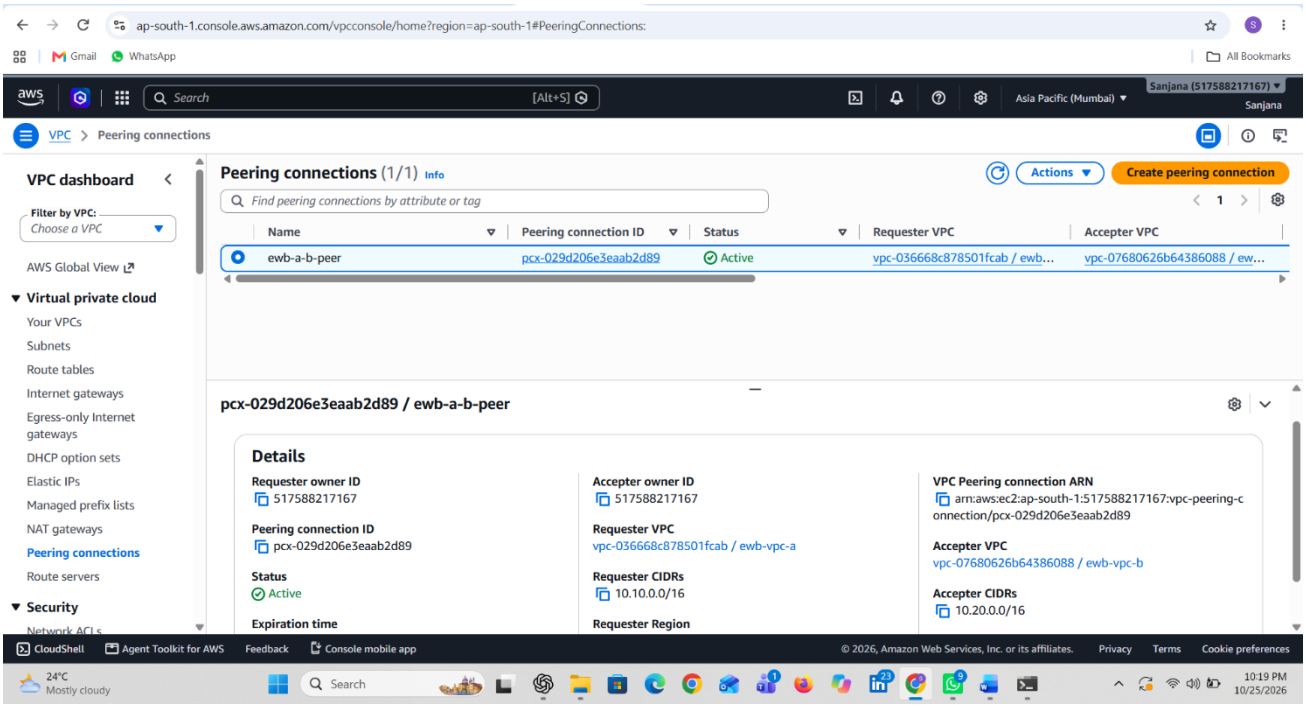


Figure 3: Public Route Table Configuration for VPC A

The ewb-rt-a-public route table was associated with the ewb-a-public subnet (10.10.1.0/24). It contained the local VPC route, a default route to the internet gateway, and a route for the VPC B CIDR (10.20.0.0/16) through the VPC peering connection.

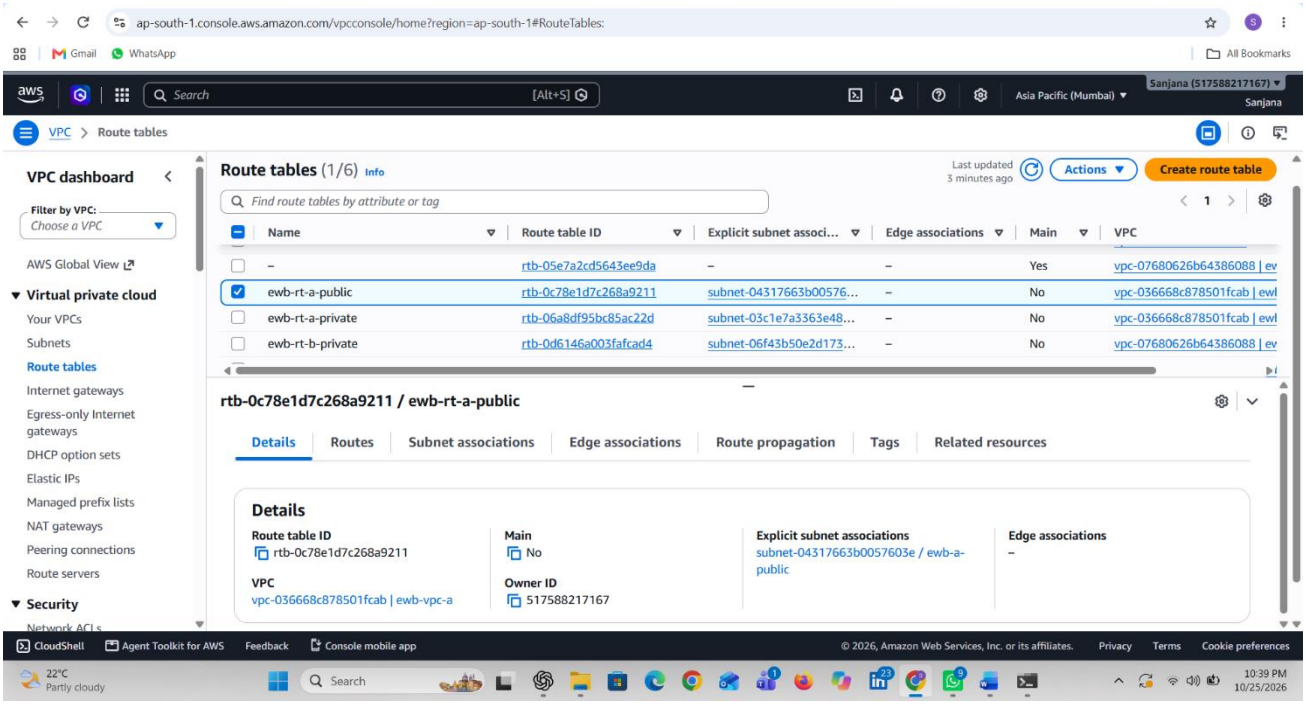


Figure 4: Private Route Table Configuration for VPC A

The ewb-rt-a-private route table was associated with the ewb-a-private subnet (10.10.2.0/24). It contained the local route for VPC A and a route directing traffic destined for VPC B (10.20.0.0/16) through the VPC peering connection.

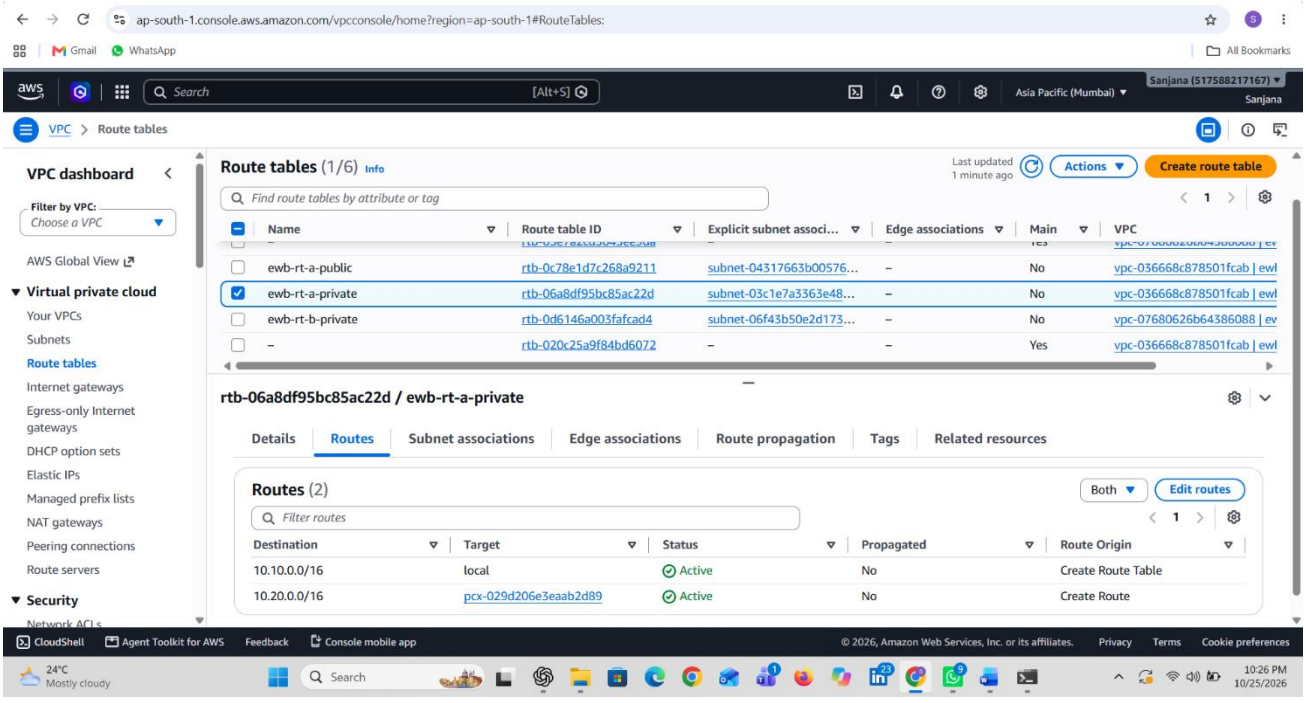


Figure 5: Private Route Table Configuration for VPC B

The ewb-rt-b-private route table was associated with the ewb-b-private subnet (10.20.1.0/24). It contained the local route for VPC B and a route directing traffic destined for VPC A (10.10.0.0/16) through the VPC peering connection, enabling return connectivity.

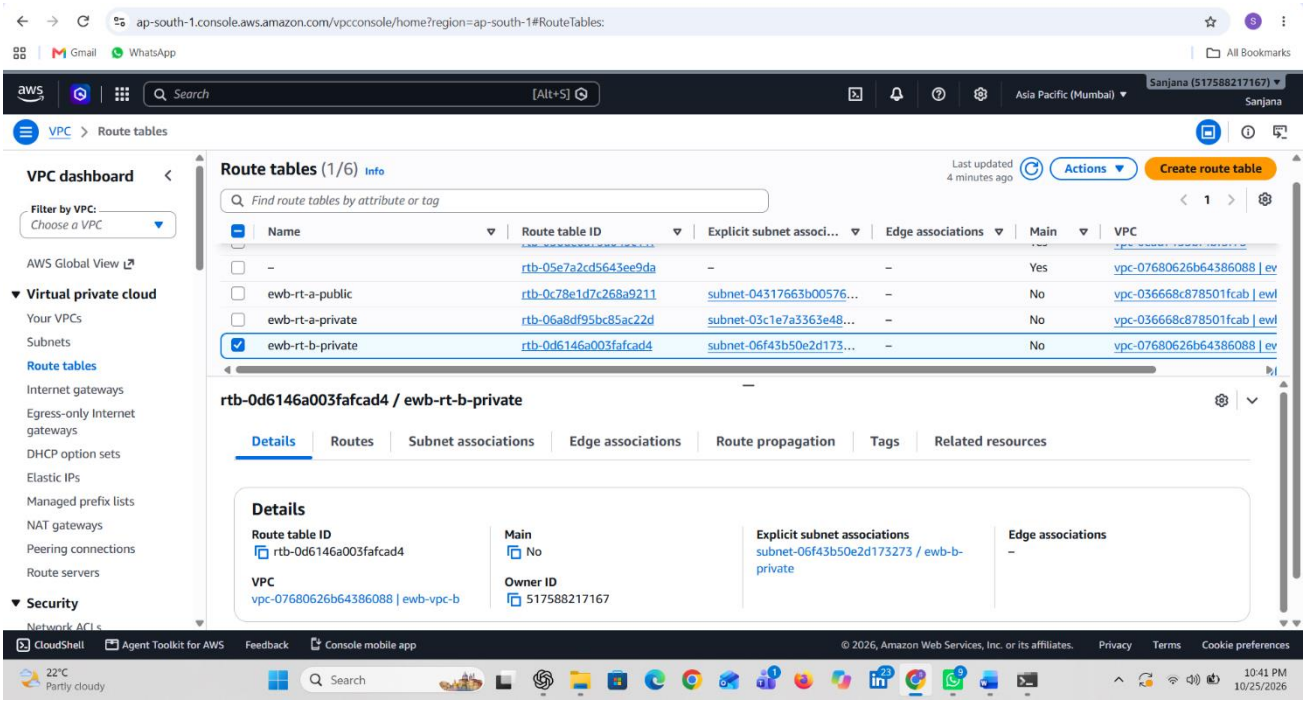


Figure 6: Inbound Security Group Rules for the Bastion Host

The ewb-sg-bastion security group allowed SSH access over TCP port 22 from the configured client public IP address using a /32 CIDR restriction.

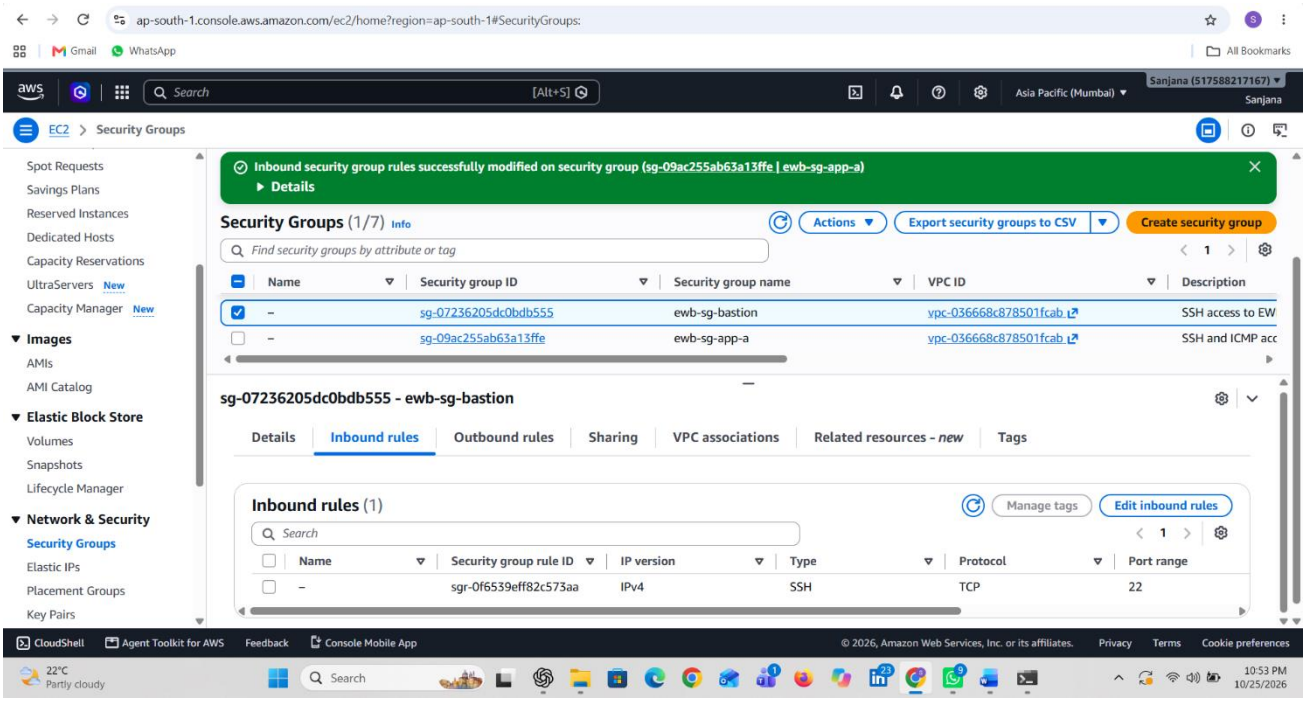


Figure 7: Inbound Security Group Rules for App-A

The ewb-sg-app-a security group allowed SSH over TCP port 22 from the configured source 10.10.1.10/32 and ICMP Echo Requests from App-B (10.20.1.10/32) for connectivity testing.

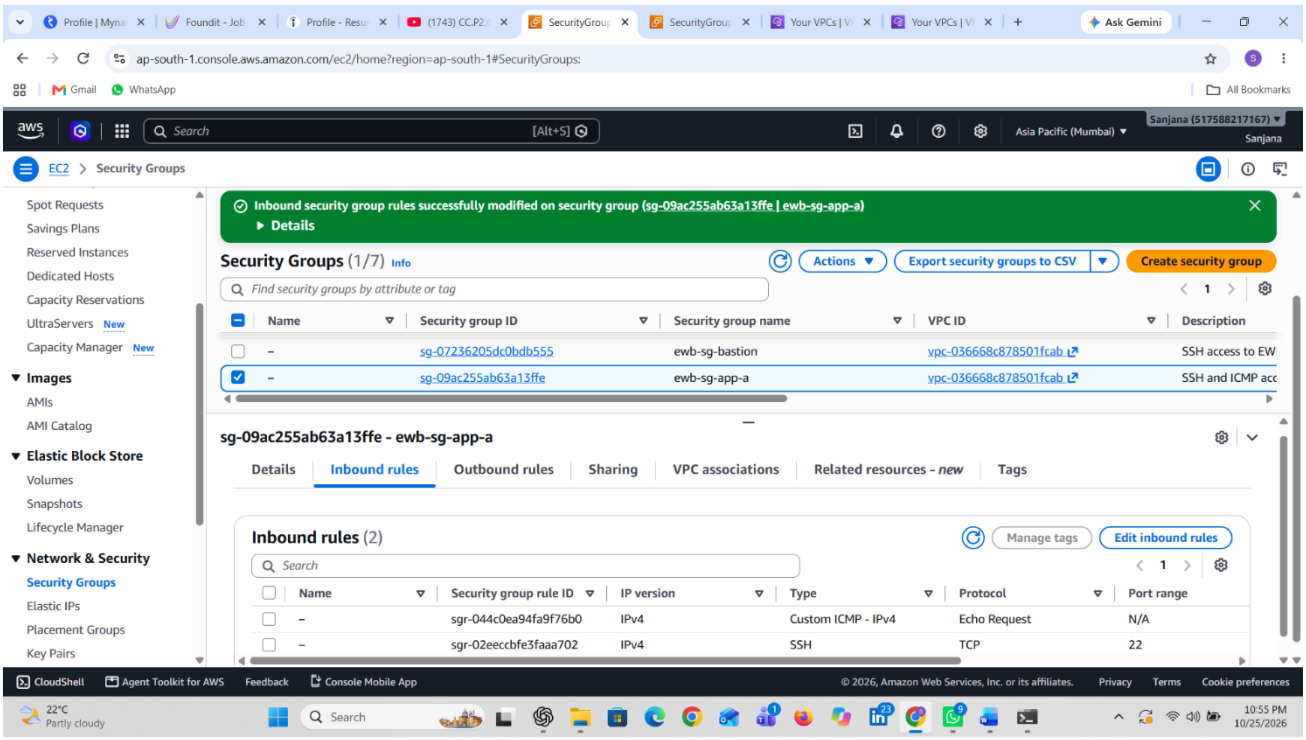


Figure 8: Inbound Security Group Rules for App-B

The ewb-sg-app-b security group allowed SSH over TCP port 22 from the configured source 10.10.1.10/32 and ICMP Echo Requests from App-A (10.10.2.10/32) for connectivity testing.

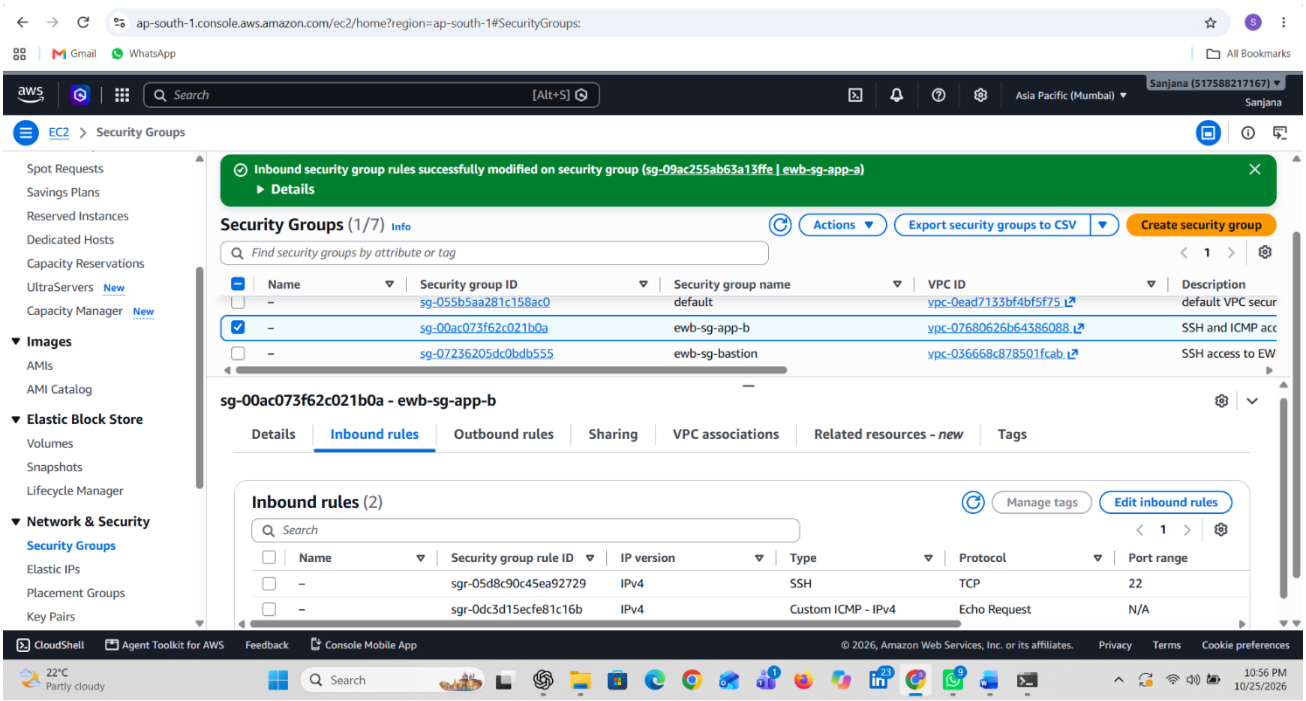


Figure 9: Successful Forward Connectivity Test from App-A to App-B

App-A (10.10.2.10) successfully sent ICMP Echo Requests to App-B (10.20.1.10) through the VPC peering connection. Four replies were received with 0% packet loss, confirming forward connectivity.

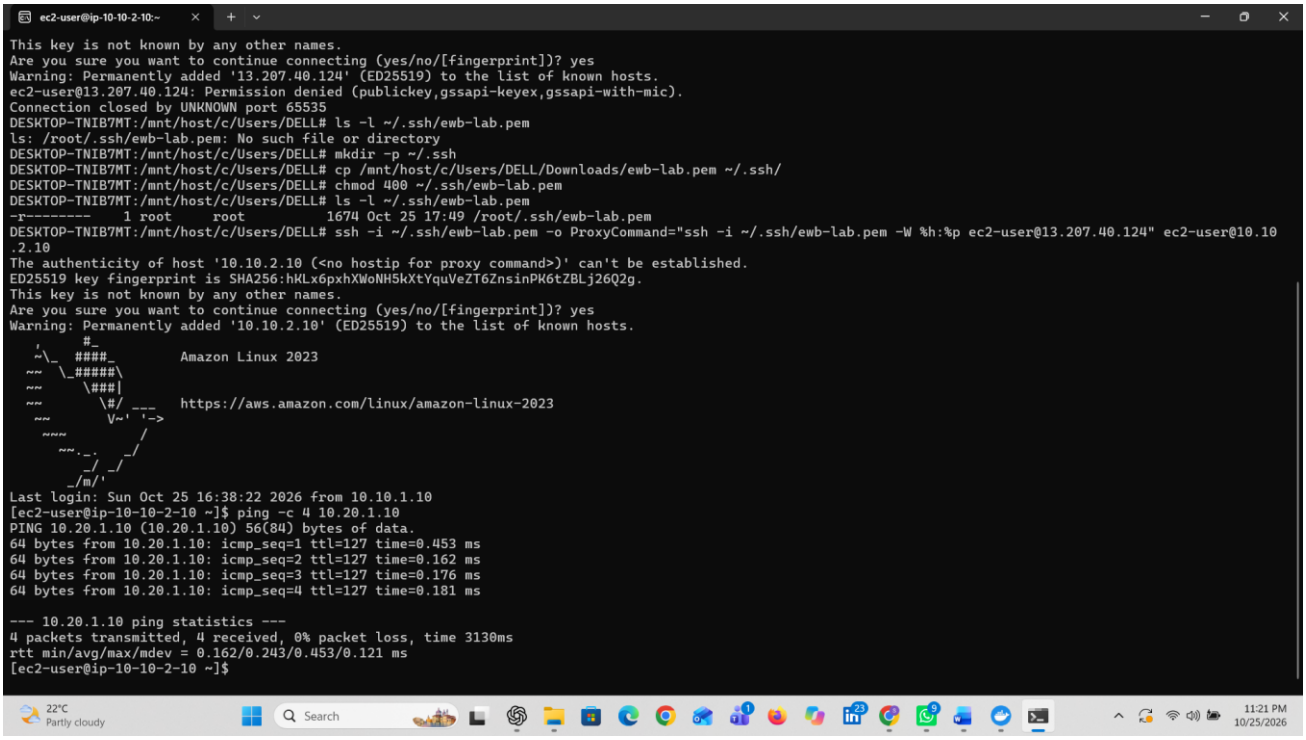


Figure 10: Successful Reverse Connectivity Test from App-B to App-A

App-B (10.20.1.10) successfully sent ICMP Echo Requests to App-A (10.10.2.10) through the VPC peering connection. Four replies were received with 0% packet loss, confirming reverse connectivity.

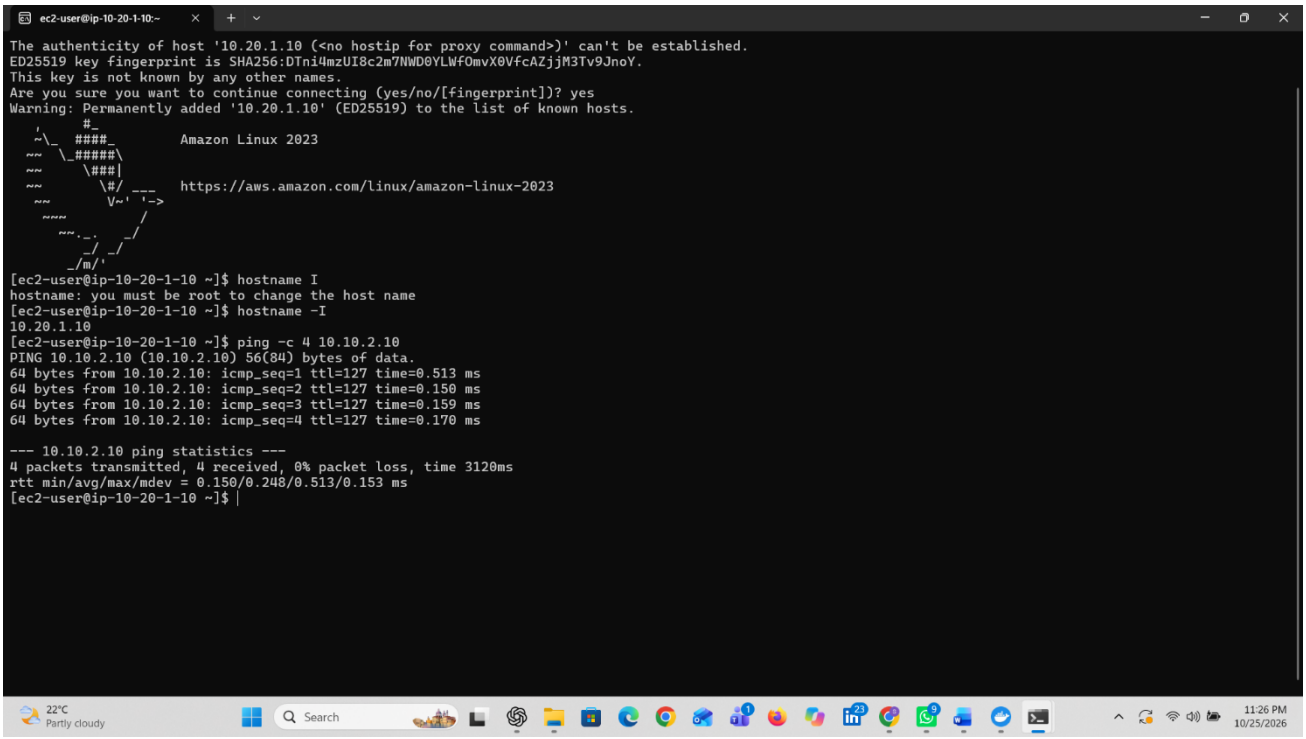


Figure 11: SSH Access to App-B Through the Bastion Host

SSH access to the private App-B instance was demonstrated through the Bastion host. App-B used private IPv4 address 10.20.1.10 and did not require a public IPv4 address for this access method.

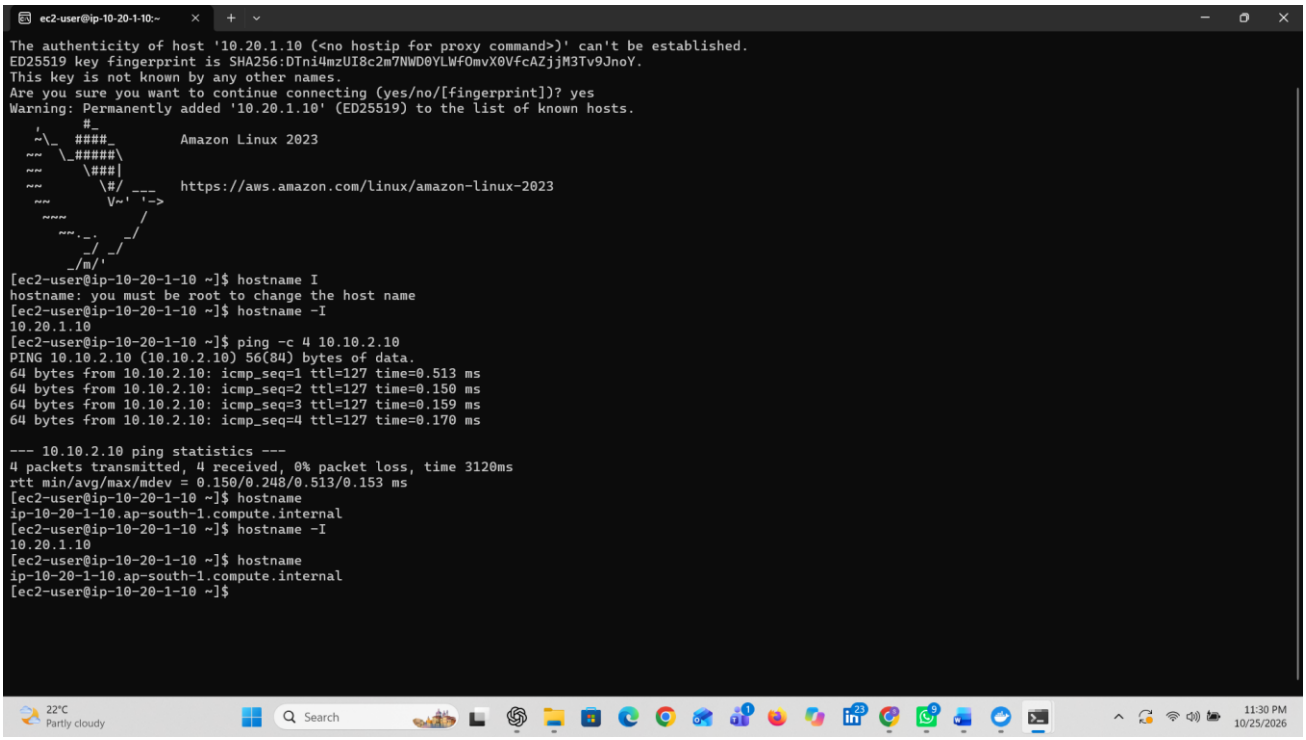


Figure 12: Controlled ICMP Connectivity Failure

The relevant ICMP security group rule was temporarily changed to block the test traffic. The subsequent ping test failed, demonstrating how security group rules control network communication between the private instances.

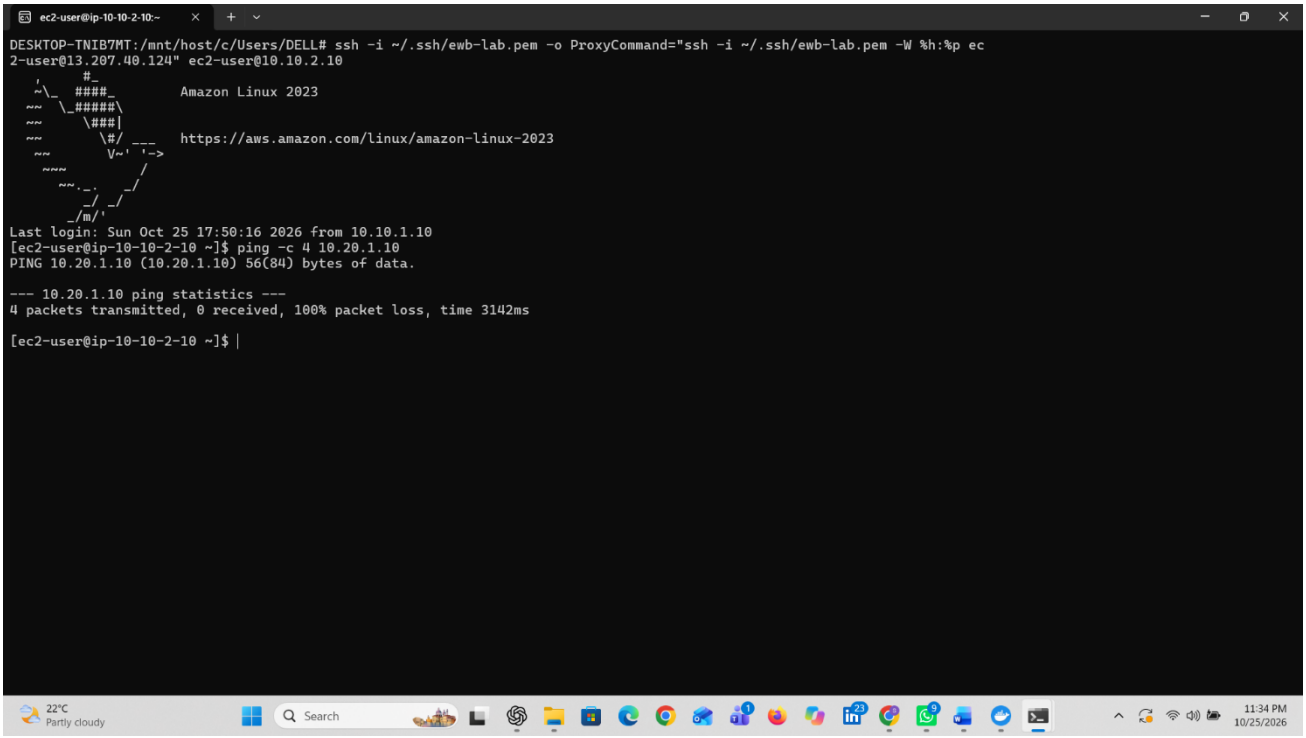


Figure 13: Successful Connectivity Recovery After Restoring the ICMP Rule

The required ICMP security group rule was restored, and the ping test was repeated. Connectivity was recovered, confirming the importance of correctly configured security group rules.

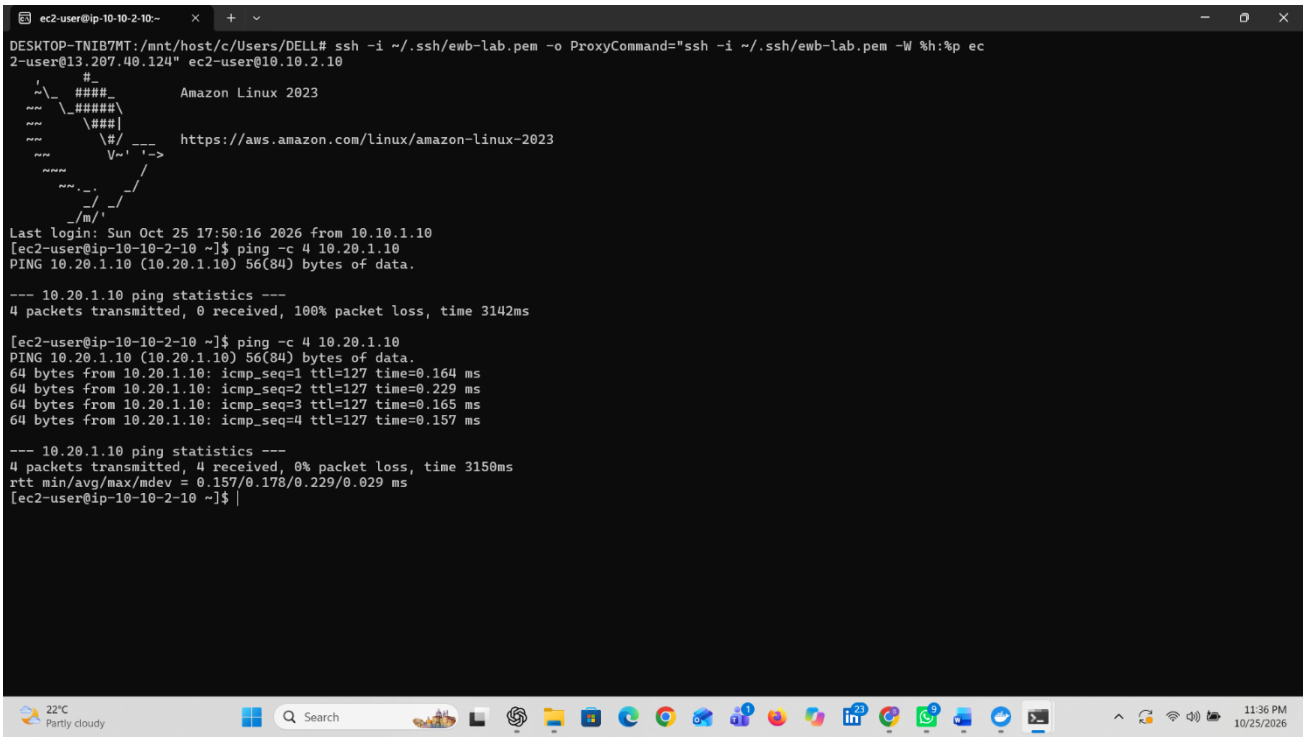
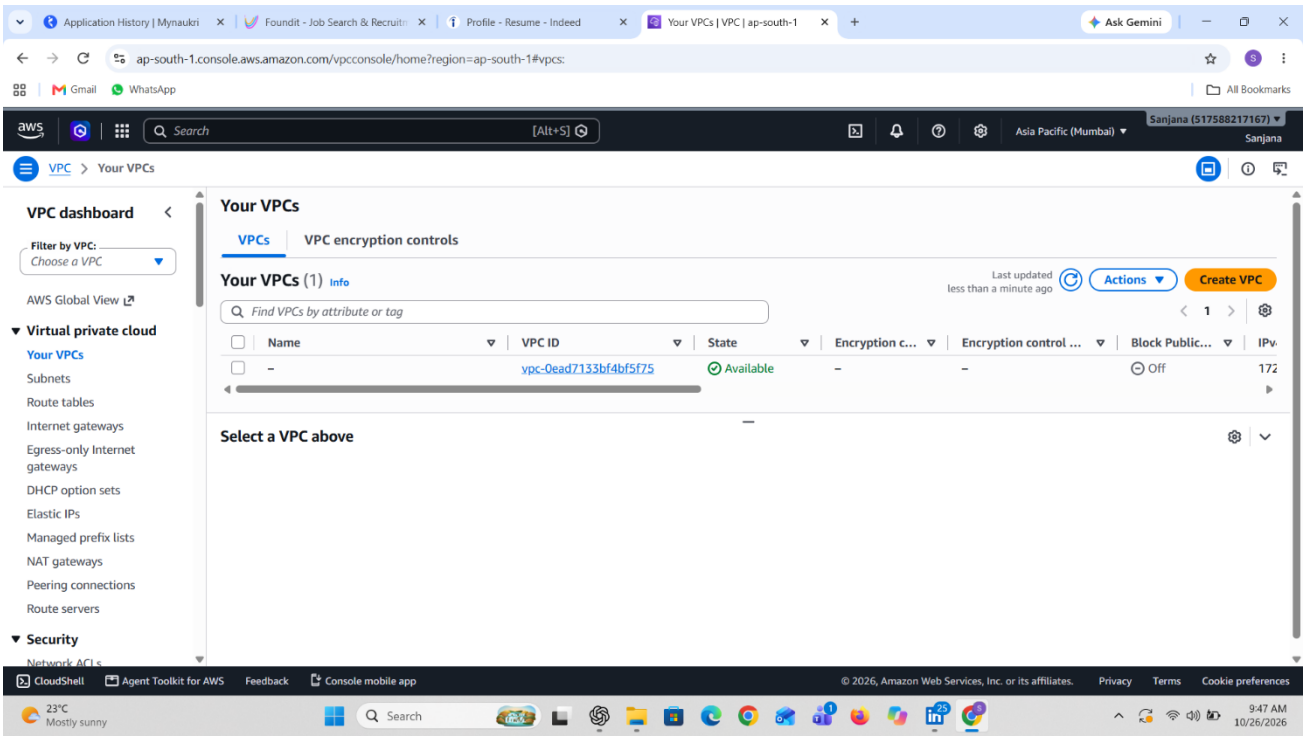


Figure 14: AWS Resource Cleanup After Project Completion

After completing the VPC peering configuration and connectivity tests, the AWS lab resources were cleaned up to avoid unnecessary charges. The EC2 instances were terminated, and the VPC peering connection, custom route tables, subnets, security groups, internet gateway, and VPCs were deleted. No leftover EBS volumes, snapshots, or Elastic IPs were found.



7. Forward and Return Traffic Path

For App-A to reach App-B, traffic originates at 10.10.2.10 in VPC A. The private route table in VPC A directs traffic for 10.20.0.0/16 to the VPC peering connection. VPC B's route table allows the destination VPC-A CIDR to return traffic through the same peering connection. The reverse test validates the opposite direction: App-B sends to 10.10.2.10 via the peering route, and VPC A has the corresponding return route for 10.20.0.0/16.

Successful connectivity depends on all required parts working together: active peering, routes in both directions, and security-group rules that permit the test traffic. VPC peering does not automatically add routes to route tables.

8. Cleanup

After collecting test evidence, the lab resources were cleaned up. The inventory records that all three EC2 instances were terminated; no leftover EBS volumes, snapshots, or Elastic IPs were found; and the peering connection, custom route tables, subnets, security groups, internet gateway, and both VPCs were deleted.

Cleanup checklist

• EC2 instances terminated.

• No leftover EBS volumes or snapshots found.

• No Elastic IPs found.

• VPC peering connection deleted.

• Custom route tables, subnets, security groups, and internet gateway deleted.

• Both VPCs deleted.

9. Learning Summary

This project provided practical experience in designing isolated VPC networks, planning non-overlapping CIDR ranges, placing a bastion host in a public subnet, keeping application instances private, and using VPC peering for private communication. It also demonstrated why route tables must contain forward and return routes and why security-group rules must allow the intended traffic. Testing both directions, deliberately blocking ICMP, restoring connectivity, and cleaning up resources reinforced the importance of validation and cost-aware cloud lab management.